

Privacy-Preserving Emotion Recognition: User-Controlled Sharing and On-Device Learning Midterm Report)

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Abstract—As digital interactions become increasingly central to daily life, there is a growing need for machines to recognize and respond to human emotional states in order to optimize user experience. However, emotions are nuanced and complex, with situational variables (appraisal, environment, context, timing) impacting the development of emotions; a discrete input cannot be used to evaluate the intricate tapestry that is a human’s emotional state. This research identifies a way to measure and classify multiple facets of human input into predictive emotional states to help identify education and learning-related emotions, to create a robust dataset of facial expressions that will hopefully help guide future machines to help identify a user’s state while learning, and create an optimized learning model to fit their needs.

Index Terms—emotion recognition, academic emotions, epistemic emotions, federated learning, privacy protection.

I. INTRODUCTION

Emotion Recognition (ER) can be defined as deducing the emotional state of an individual using given feature inputs (facial, vocal, biometric, etc). Its research is driven to improve the performance of human-centered intelligent systems since emotion is integral in understanding user interaction and satisfaction. ER has found application across a range of domains — including convergence, augmentation of human judgment, deconfounding, production, tutoring, avoidance behavior, alerting, and entertainment — where systems adapt their output upon detecting a significant emotional signal [1].

One of the many applications of ER is within the education realm. The online education industry has entered a phase of rapid expansion, with the market valued at \$10.4 billion in 2024 and projected to nearly double to \$23.7 billion by 2030, reflecting a compound annual growth rate of 14.5% from 2025–2030 [2]. This sustained growth underscores the increasing adoption of digital learning platforms and the rising demand for flexible, accessible educational alternatives worldwide. Within this context, emotionally sensitive tutoring systems offer a mechanism for assessing instructional effectiveness in real time, allowing a system to adjust its teaching approach to better serve the individual learner.

Despite this potential, the literature on emotion specific to learning contexts remains limited. Existing studies predominantly rely on variants of Ekman’s six basic emotions, a

taxonomy developed for general affective expression rather than for the cognitive states characteristic of learning, and which may introduce unnecessary complexity when applied to an educational context. This research aims to address that gap by collecting and developing a labeled dataset specifically targeting emotion recognition in education.

The collection of biometric and facial data, however, introduces privacy risk. Centralized systems that store raw biometric data on a server increase exposure to data breaches, unauthorized access, and surveillance misuse, and they introduce a single point of failure: a compromise of the central server jeopardizes the entire dataset [3]. Accordingly, this research must not only support varying levels of user data disclosure, but also identify the factors that influence users’ disclosure preferences.

II. LITERATURE REVIEW

In the following sections, we use relevant literature to lay the groundwork of our research.

A. Privacy Protection for Users’ Facial Information

New methods have been developed to be able to give users more protection and autonomy over their privacy. Rather than transmitting a user’s unprocessed data, systems can instead extract and share only the information relevant to emotion detection, at varying levels of abstraction. Three categories of derivable data are relevant here: Raw data: the unprocessed video itself — the actual sequence of image frames capturing the user’s face, surroundings, lighting, and anything else in view. It is the richest and most sensitive form, since it contains identifiable information (who the person is, where they are) far beyond the emotional signal of interest. Facial data: the intermediate representation extracted from raw frames: facial action units, landmark coordinates, head pose, and gaze direction. The raw imagery is discarded, leaving numerical descriptors of facial movement. This strips away much identifying context (background, full appearance) while retaining the information needed to infer expression, though some features like landmark geometry can still be partially identified. Parameter data: the weights (and gradients) of the machine-learning model after it has been trained on a user’s facial features locally. No

images or feature vectors leave the device; Only the learned parameters that encode patterns across the data.

Parameter data forms the basis of federated learning, an architecture in which model updates, rather than raw data, are transmitted to a central server. This allows users to selectively share only the minimal information necessary for emotion inference, rather than exporting raw video.

B. The Complexity of Emotions

Prior to addressing the technical challenge of identifying, recording, and labeling emotion, it is necessary to establish a working definition of the term as used in this study. "Emotion" may be understood in either a narrow or broad sense: narrowly, it denotes an intense and dominant affective state that directly influences decision-making and behavior; broadly, it denotes a subtler influence that colors thought and action without necessarily dominating it, except when sustained over time. This distinction builds upon established theoretical frameworks — including Ekman's six basic emotions, Plutchik's wheel of emotions, and the valence-arousal model — which collectively suggest that emotion is more accurately understood as a continuous spectrum rather than a fixed set of discrete, universal categories [1].

Regardless of classification methods, emotion cannot be fully captured by any single set of discrete features. Numerous factors contribute to the experience of an emotion, including physical embodiment, somatic arousal, situational appraisal, and contextual influence, among others [1]. Because this study aims to measure and label emotional states for the purpose of training a predictive model, the resulting labels must be representative and resistant to overgeneralization or miscategorization. Given the constraints of a 10-week research timeline, however, this study adopts a narrowed scope: rather than attempting to capture the full space of learning-related emotion, it focuses on accurately eliciting and labeling a small set of emotions identified as most essential to the learning process.

C. Academic Emotions

To establish a structured understanding of emotion within learning, this study turns to the work of Reinhard Pekrun, whose research on achievement emotions and control-value theory has substantially shaped the modern understanding of how emotion operates within academic contexts. Pekrun's framework offers a taxonomy of academic emotion, organized into four primary categories: achievement, epistemic, topic, and social emotions. This taxonomy provides the foundation from which the present study identifies its target emotion set.

Achievement emotions are tied to perceptions of success and failure (e.g., anxiety preceding an exam, pride following a strong grade). Epistemic emotions are triggered by the information itself (e.g., curiosity in response to a surprising fact, confusion when material fails to align with prior understanding). Topic emotions are evoked by the subject matter being studied (e.g., empathy toward a character in a novel, excitement about a scientific concept). Social emotions arise

from interpersonal relationships within the academic setting (e.g., gratitude, envy, or admiration directed toward instructors or peers). [4]

D. Selection of Epistemic Emotions

This study focuses on epistemic emotions for two primary reasons. First, epistemic emotions are less dependent on external context than the other three categories, making them more tractable to elicit in a controlled HCI study. Achievement and topic emotions require accounting for individual preferences and contextual variation that differ widely across participants, complicating standardization. Social emotions present a similar challenge, as they depend on factors such as societal norms, hierarchical dynamics, and personal schema that are difficult to control or replicate in an HCI setting. Epistemic emotions, by contrast, can be organically and reliably elicited in the moment, requiring nothing more than a cognitive misalignment between a learner's expectation and the presented information [4].

Second, epistemic emotions are more closely associated with deep learning outcomes. Achievement emotions are driven by performance goals — appraisals of perceived competence and the value placed on succeeding — whereas epistemic emotions are driven by mastery goals: the motivation to resolve cognitive incongruity that arises when new information surprises, contradicts, or challenges a learner's existing beliefs [5]. Performance goals tend to promote extrinsically motivated behavior, such as memorization or gaming the assessment system, while mastery goals foster enjoyment and initiative, reducing the likelihood of boredom and encouraging deeper engagement with material.

E. Narrowing the Epistemic Emotion Set

Because the category of epistemic emotion encompasses a wide range of affective states, further narrowing was required to identify a subset that could realistically be elicited within a single HCI session. A review of theoretical and experimental literature on epistemic emotions in learning contexts identified surprise, curiosity, confusion, and boredom as the most prevalent and well-supported states [6]-[10]. Pekrun notes that learners' enjoyment is tied to task value — material perceived as interesting and personally relevant tends to produce enjoyment, while material lacking such value tends to produce boredom [5].

D'Mello and Graesser identify a closely related set of learning-centered affective states, grouping surprise and curiosity under a unified category termed flow or engagement [11]. Critically, their model further proposes that these affective states do not occur independently of one another; rather, certain states function as transitional links that a learner must pass through to reach others. Specifically, the model holds that flow and confusion can transition into one another, confusion and frustration can transition into one another, and frustration and boredom can transition into one another, forming a connected chain of affective states rather than a set of isolated emotional categories.

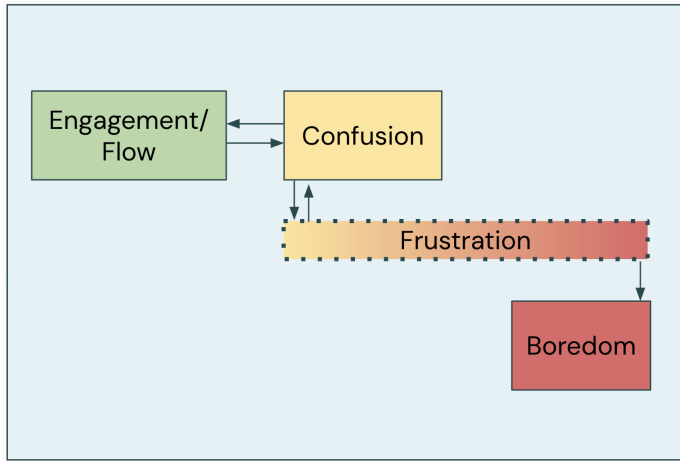


Fig. 1. Main Affective Learning States

III. PROJECT DESCRIPTION

Based on the literature reviewed above, this study focuses on eliciting three core affective states in participants: confusion, flow/engagement, and boredom. The objective is to construct a dataset of labeled facial expression data sufficient to train a machine learning model capable of identifying these states in new learners. The resulting model is intended to detect confusion in a student and distinguish between productive confusion (engagement with a soluble cognitive challenge) and unproductive confusion (a stalled or unresolved impasse), and to adjust its teaching strategy accordingly: maintaining pace when a learner exhibits flow or productive friction, and modifying instructional approach when a learner shows signs of disengagement or boredom. Throughout this process, the study will assess and accommodate participants' privacy preferences to ensure the safe and consented use of recorded data.

IV. METHODOLOGY

A. Survey Design

To better understand the factors that shape users' decisions around information disclosure, this study designed a four-part survey, informed by the survey structure used in Dehling et al.'s study of internet users' willingness to disclose biometric data [12]. The survey consists of the following stages: Baseline Rating: Rate without prior knowledge of the available options Education: get educated about data types and how they work Comprehension: Take a short comprehension quiz (failure = excluded) Trait Rating: Re-rate baseline questions across all 3 data types

For each data type, participants rate the following on a 1–100 scale: willingness to share, perceived response efficacy, and perceived loss of privacy due to disclosure. This design accounts for the issue of incomplete and asymmetric data identified in prior survey-based disclosure research, ensuring that participant responses reflect informed rather than naive judgments.

B. Game design background

The elicitation task is designed as a game-based interaction, motivated by D'Mello's finding that confusion, when appropriately induced, can be beneficial to learning rather than purely disruptive. Prior work has demonstrated this by presenting learners with information at varying levels of difficulty or ambiguity, then assessing comprehension through embedded quizzes or exam-style checks, and pairing performance scores with self-reported emotional state to establish ground-truth labels for the emotion experienced at that moment [13].

C. Data collection and classification

Through the gamified design, the system intends to collect three modalities of data: raw data, facial features, and model weights. As the user plays the game, the app records and collects facial and audio files as raw data. These files are then passed into OpenFace for feature extraction, a software toolkit designed for automatic facial behavior analysis. OpenFace outputs a comma-separated-value (CSV) file containing 538 variables for each frame, including facial action units, gaze, and head pose estimations. For model development, the project uses the CREMA-D dataset, a collection of 7,442 clips from 91 actors and actresses with diverse age and ethnicity, expressing six emotions: happy, sad, anger, fear, disgust, and neutral [14]. The files from the dataset are synthesized into a feature matrix, where each row represents one video clip and each column represents the mean and standard deviation of each Action Unit variable, along with the emotion label. The data is split 80/20 for training and testing respectively, a common standard in Machine Learning. The current model is a Support Vector Classifier (SVC) from the scikit-learn library, trained to predict the emotion label given all AU features. The parameters are as follows: kernel = 'rbf', C = 1, and gamma = 'scale'. The RBF kernel was selected based on the recommendation of Dr. Cao, while all other parameters are scikit-learn defaults used as a baseline. Currently, the model yields an accuracy of 60.6%. The final step of the pipeline sends the model weights to a central server, which aggregates weights from all users' devices and returns an updated model to each app.

V. CONCLUSION

This research has laid the theoretical and methodological groundwork for constructing a labeled facial-expression dataset targeting epistemic emotions relevant to learning. Several next steps remain before data collection can begin. The IRB proposal is currently pending submission, and approval is required prior to any participant-facing study activity. In parallel, the specific questions and content for the elicitation game — covering both comprehension assessment and emotional labeling — must be finalized to ensure the task reliably induces the target affective states. Development of the facial feature extraction pipeline using OpenFace will continue, with particular attention to action unit (AU) analysis as the basis for downstream classification. Finally, this research will begin investigating and developing the multimodal application

intended to integrate survey data, game-based elicitation, and facial expression classification into a unified system. Together, these next steps will move the project from theoretical framing toward empirical data collection and model development.

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